

# Hossam Elsherbiny

## ML Engineer

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### Professional Summary

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ML Engineer with hands-on experience building and deploying machine learning systems for time-series forecasting, NLP, and computer vision. Skilled in Python, TensorFlow, and OpenAI tools, with a strong record of delivering real-time AI applications. Experienced in optimizing models, integrating APIs, and developing end-to-end ML pipelines to solve practical problems with measurable results.

### Experience

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**ML Engineer**, Acrow – Cairo Apr 2025 – now

- Extracted and processed data from Oracle databases to support time-series forecasting for over 3,000 SKUs.
- Designed, trained, and evaluated forecasting models (SARIMA, XGBoost, CatBoost, Prophet, LightGBM, and neural networks), lowering forecast error from 80% to 55%.
- Engineered an intuitive GUI-based executable application to automate the end-to-end data extraction, model training, and forecasting pipeline, decreasing the manual process time from 3 days into 1.5 hours.
- Partnered with cross-functional teams to validate model outputs and integrate forecasts into the company's planning processes, boosting profit by over 10.5%.

**Data Scientist Trainee**, DEPI Apr 2024 – Oct 2024

- Conducted exploratory data analysis and contributed to the development of machine learning models to generate actionable insights.
- Partnered with colleagues to design and implement scalable, data-driven solutions.

**AI Trainee**, ITI July 2024 – Sep 2024

- Completed coursework in exploratory data analysis, statistics, probability, and linear algebra.
- Acquired foundational knowledge of machine learning algorithms, including core concepts of convolutional and recurrent neural networks.

### Projects

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#### EnergiQ Platform

- Deployed an end-to-end, OpenAI-powered AI agent to address user inquiries about electric vehicles, charging stations, and related topics, serving 7,000 users.
- Implemented a function-calling framework to interface with the backend database, enabling retrieval of user-specific metrics (e.g., last session details, monthly spending), attained  $F1 = 0.94$  of Intent Accuracy.
- Enhanced a Retrieval-Augmented Generation (RAG) system to answer general platform, EV, and charging-related queries using internal documentation, achieving  $\text{Recall@3} = 0.95$ .
- Enabled voice chat feature using OpenAI APIs (whisper-1, tts-1), reduced inference latency to 78 ms/token.

#### Knowva Platform

- Designed and implemented an AI-driven language learning platform that assesses user proficiency through an adaptive exam and personalized content targeting 500k users.
- Integrated OpenAI's Whisper model for real-time speech-to-text conversion, enabling seamless voice input processing and level evaluation with  $\text{WER} = 0.232$ .
- Optimized the Gemini model to generate personalized responses and custom learning materials, secured a tool response rating of 4.64/5 from users.
- Employed xTTS for text-to-speech functionality and WAV2Lip for lip-sync synchronization, enhancing listening comprehension and pronunciation skills, having 0.92 Cosine Speaker Similarity.

## Brain Tumor Detection

- Curated and augmented imaging datasets to optimize training and enhance classification performance.
- Utilized EfficientB3Net as the foundational model via transfer learning to classify brain tumors into 3 categories.
- Attained approximately 98.7% accuracy on the test set by fine-tuning the model on medical imaging data.
- Built an interactive Streamlit interface to enable instant predictions and facilitate user interaction.

## Spacecrafts Reuse Feasibility

- Spearheaded data extraction from SpaceX APIs to support spacecraft reuse feasibility analysis.
- Cleaned, transformed, and structured datasets to ensure accuracy and readiness for analysis.
- Extracted key patterns using EDA and applied data visualization to identify key trends and relationships among features.
- Evaluated various machine learning models (KNN, SVM, Decision Tree, Random Forest) accomplishing 92% accuracy and selected the optimal classifier to determine spacecraft reuse potential.

## Skills

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**Programming Languages:** Python, SQL, R

**Analytics Frameworks:** Numpy, Pandas, Scipy, Matplotlib, Seaborn

**ML Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, Prophet

**NLP Tools:** OpenAI, Gemini, Hugging Face Transformers

**CV Tools:** OpenCV, Streamlit, WAV2Lip

**Tools:** Git, Docker, Jupyter, Tableau, PowerBI, Spreadsheets

**Databases:** SQLite, PostgreSQL, Oracle

## Education

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**Bachelor's Degree in Computer Engineering**

Oct 2020 – Jul 2025

Zagazig University

## Volunteering

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**IEEE Zagazig Student Branch**, Computer Society Vice Chairman

- Led a team of 35 novice programmers, instructing them in core algorithms and data structures.
- Mentored participants to successfully complete Level 0 challenges from the ICPC Assiut problem set.
- Contributed to organizing the Mutex event and created problem sets for the Semaphore contest.
- Prepared and coached team members for competitive programming events, including ICPC and IEEEExtreme.

## Awards

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**Second Place, Banha Hackathon** (AI Solutions for Language Learning)

Built an AI-driven platform to assess and enhance language proficiency through personalized content and interactive practice.

**Third Place, MATE ROV Regional Competition** (Co-Pilot & Programming Lead)

Responsible for developing the vision system and implementing image processing algorithms to support underwater task execution.